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Climate 410, Lab 2
Due Monday, October 12th at 5pm

Methodology

My approach to coding this script was generally very repetitive. The focus of this lab was to solve various sets of differential equations in order to practice both understanding the concepts behind these equations and the techniques to actually solve them using code. We were given three different sets of differential equations: the Lotka Volterra equations for both species in competition for resources and predator and prey species, and then the Lorenz equations for atmospheric fluid motion.

To go about solving these equations, I compared two techniques: Euler's method and the rk45 method. This meant writing two functions to be adapted later for other contexts. For a previous homework assignment, we wrote pseudocode to describe a Python function that solved a differential equation. I was able to translate that to actual code and modify it so it could be applied to these purposes. This meant defining a certain time step, a time range, an array that contained the two species populations, and the constants in the equation. These are the parameters for the function lv_euler( ). This function uses the parameters and plugs them into the Lotka-Volterra equations given in the lab manual and returns the total time taken, a list of each time with each time step, and a list of the two populations at each time step. With those variables returned, I was able to produce graphs, as shown throughout Tasks 1 and 2. I also compared this method to the rk45 method, a professional-grade ODE solver. The instructions to build this function (lotvolt1( )) were mostly given in the lab manual, and from there I was able to make graphs of these results as well, using the same parameters as I did in lv_euler( ). These graphs are also shown in Tasks 1 and 2, and the comparisons of the two methods are analyzed there as well.

To make the functions that solve the Lotka-Volterra equations for the predator and prey populations, I followed almost exactly the same procedure as for the competitive species, except this time I used the corresponding equations also given in the lab manual. This is the function pred_prey( ). The same comparisons were made as well between Euler's method and the rk45 method. The only major difference is that this procedure included an additional task: create a phase diagram to investigate the behavior of the populations with time. The way to make a phase diagram is to plot the predator population against the prey population rather than either of them versus time. This graph is attached below in Task 3 as 24.

Once the functions were made and graphs were produced using the coefficient values given in the lab manual, I then varied some of those coefficients to see how the populations would behave with different conditions imposed upon them. I made more graphs for each of the results, and those images with their analyses are attached below in Tasks 1-3.

Finally, I analyzed the Lorenz equations for weather systems in the atmosphere using the rk45 solver method. Once again, I was largely able to copy and paste from the previous tasks to accomplish this, aside from renaming variables and changing the equations and parameters. Once

the necessary modifications were applied, the function lorenz( ) was able to be used to analyze the Lorenz equations. I analyzed results for both an r value of 23 and an r value of 25, and each of those r values also had corresponding results simulating what would happen if a perturbation was added. So, in total, the function lorenz( ) was employed four separate times in order to produce graphable results. These graphs and analyses are shown in Task 4.

An important note: in my code, the lines where I produced plots for the results are all commented out for convenience in the code writing and testing process. To see the graphs, uncomment the relevant lines as indicated in my code manuscript.

Task 1

Figure 1 below depicts the results of using both Euler's method (which we developed in HW3) and the professional-grade rk45 ODE solver to solve the Lotka-Volterra equations for two species competing for the same resources. The first image uses a time step of 1 year, as originally given in the lab handout. I then tried out varying time step values, ranging from 0.1 to 3 years. Figures 2-6 show the graphed results side-by-side for each of those time step values. The graphs plot the fraction of the capacity of the species populations versus time in years.

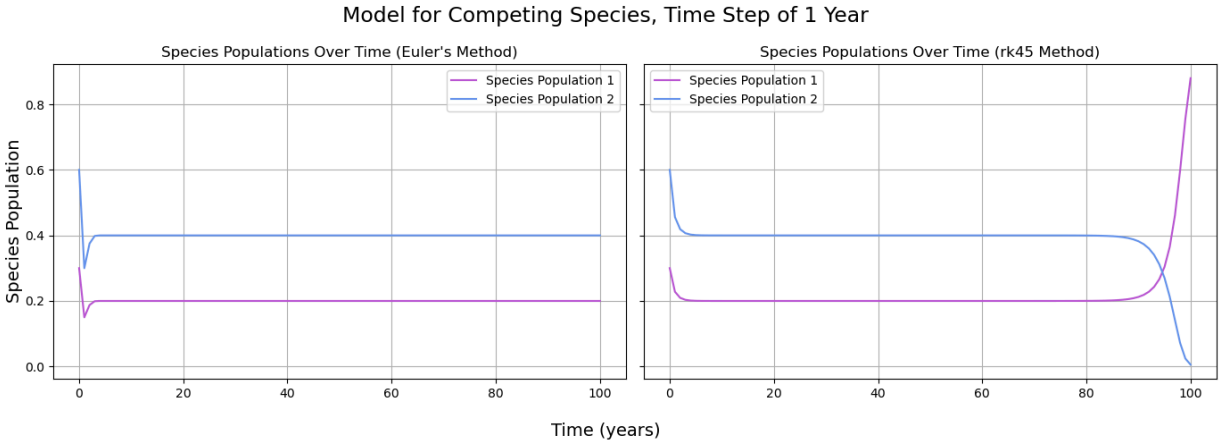


Figure 1

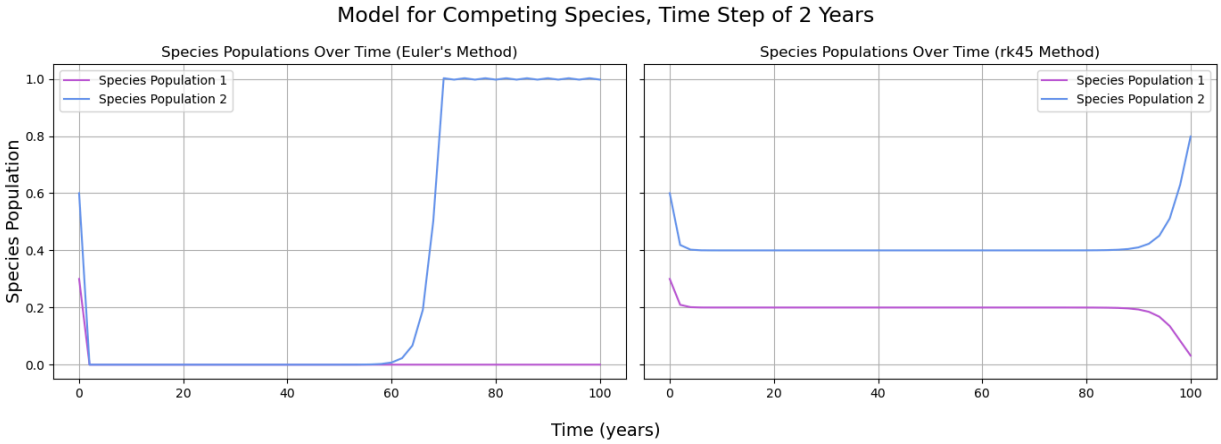


Figure 2

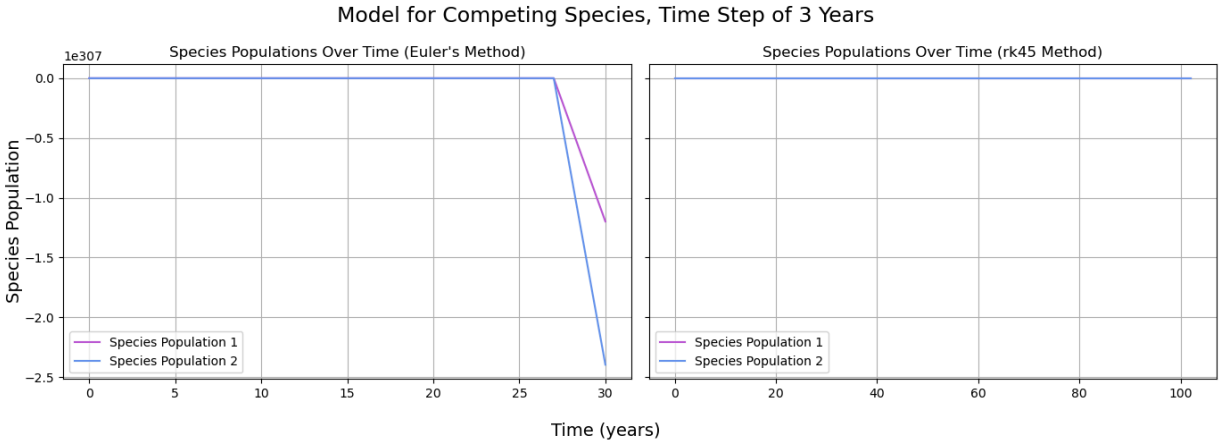


Figure 3

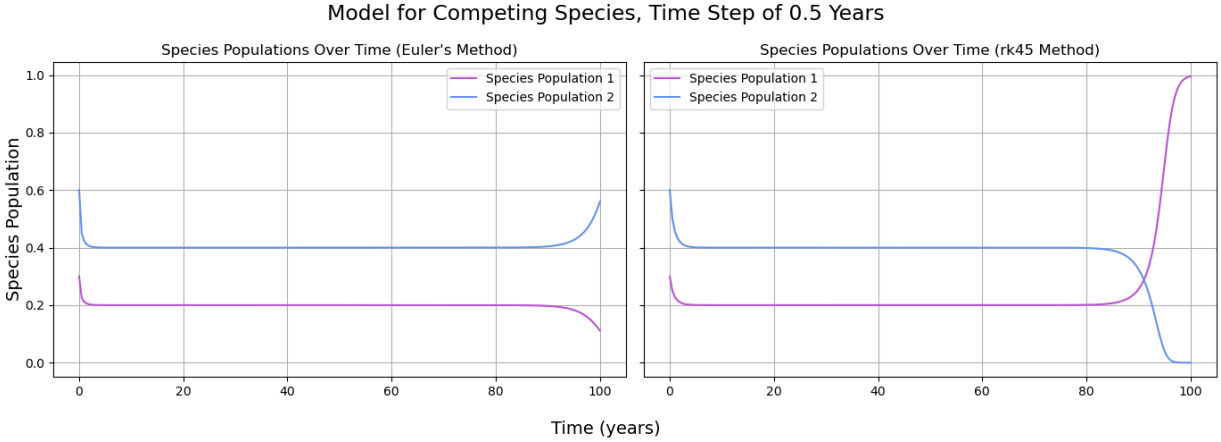


Figure 4

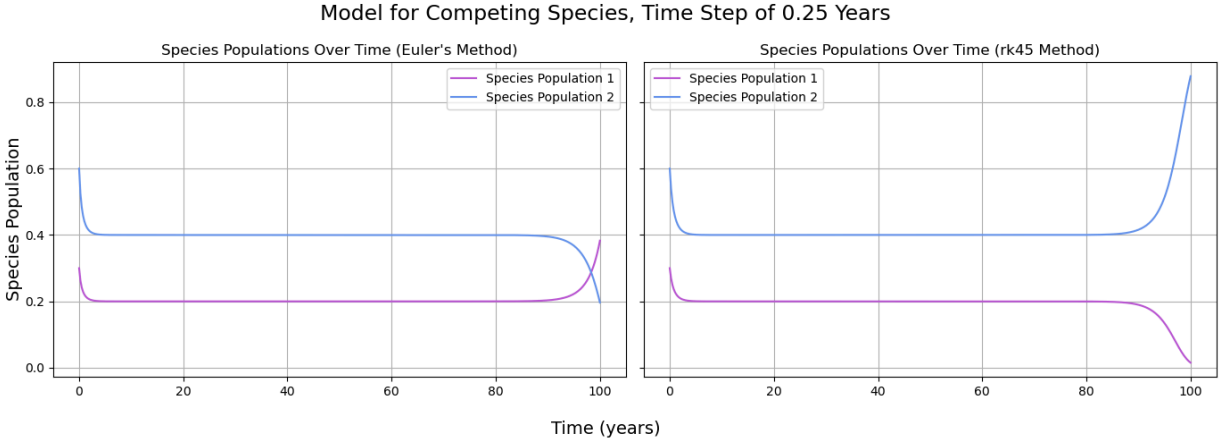


Figure 5

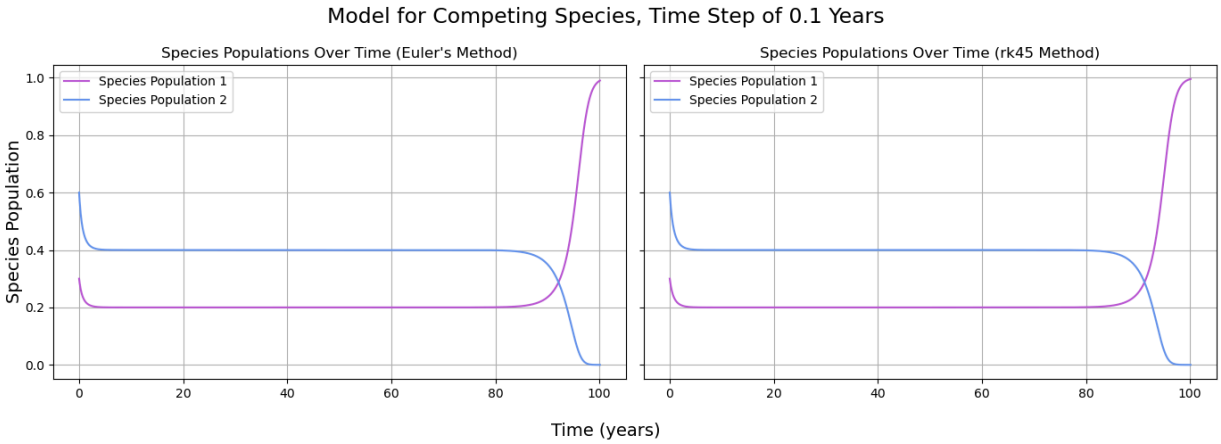


Figure 6

With the time step values I tested, it is apparent that smaller time steps result in more similar-looking graphs between the two methods. I started out with a time step of 1 year (Figure 1), and the two methods look quite different. While they both remain steady around 0.2 and 0.4, Euler's method continues that trend, while rk45 suddenly converges around approximately 95 years and then diverges to infinity. However, as that time step is decreased (Figures 4-6), the plots begin to agree more and more on a behavior, finally matching each other in Figure 6 with a time step of 0.1 years. This graph looks very similar to the rk45 result from Figure 1, but rather than diverge to infinity, the populations appear to reach a steady state, with one species extinct in the region, and the other one at full capacity. Increasing the time step greater than 1 year yielded interesting results as well. With a 2 year time step, Euler's method shows a strange graph: both populations almost immediately drop to zero until around 60 years later, when species 2 suddenly shoots up to capacity, where it then seems to oscillate around that capacity. The rk45 method completely switches which species reaches capacity and which goes extinct. Then, with a time step of 3, the results are even stranger. Euler's method yields an absurd y-scale, and rk45 stays at 0 the entire time. It is possible that these odd results are due to errors in my code. Regardless, though, it can be said that a small enough time step can create more reliable results.

Next, the same analysis will be done for the predator-prey equations. This time, I started with a time step of 0.05 years (as given in the lab manual), and from there I varied the time steps up and down to a range of 0.01-1 years. Figures 7-11 depict the results below.

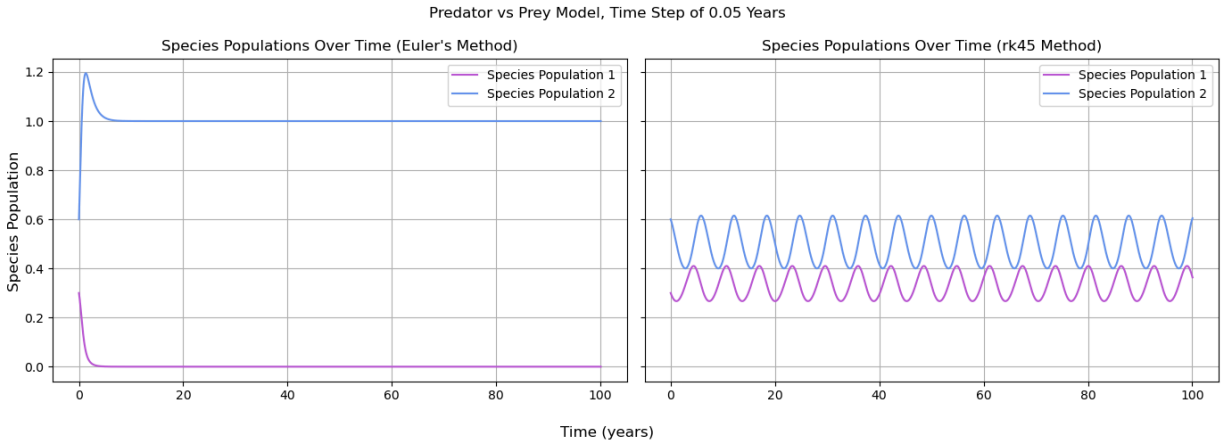


Figure 7

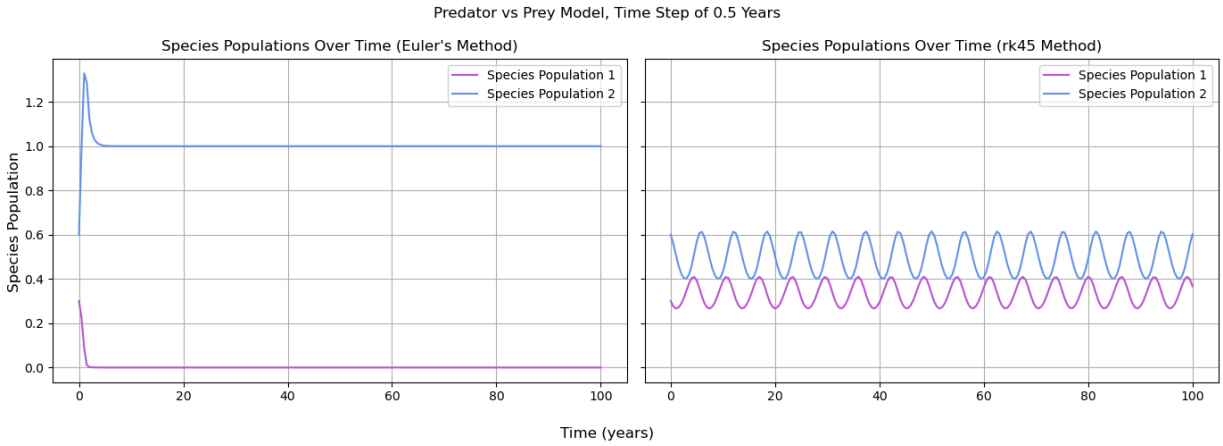


Figure 8

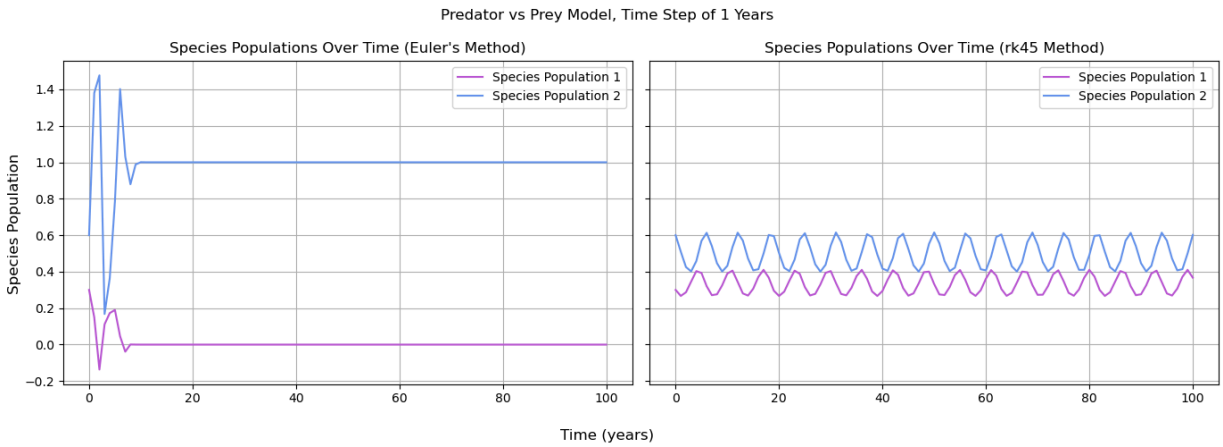


Figure 9

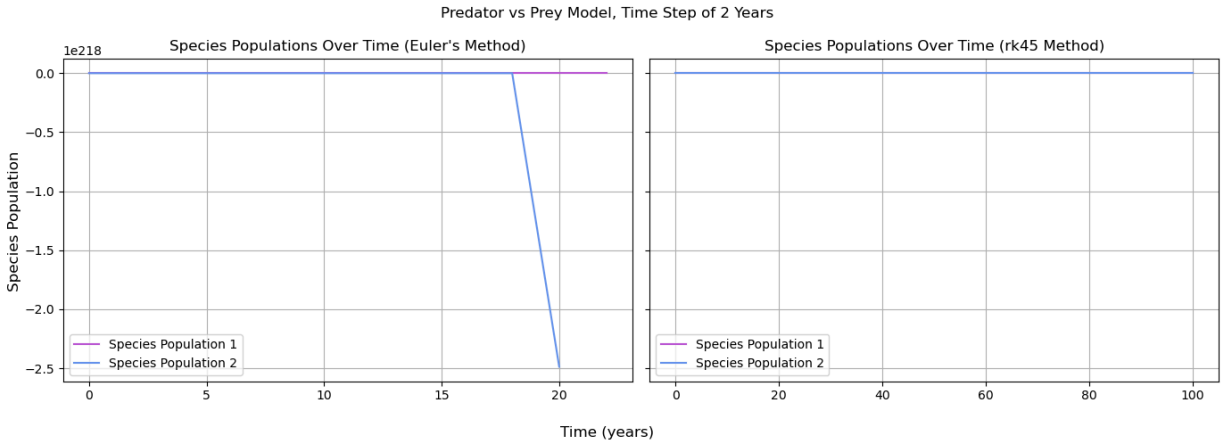


Figure 10

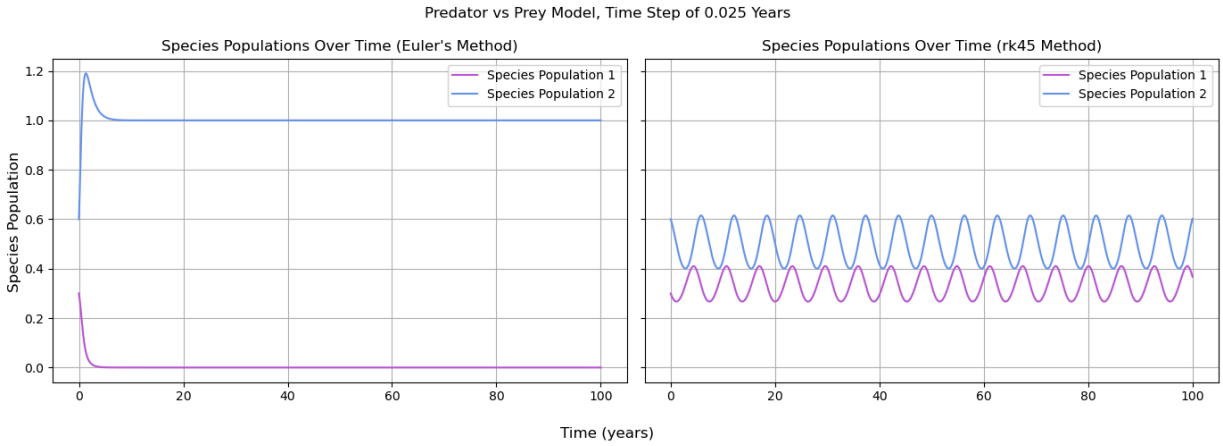


Figure 11

A similar conclusion can be drawn from these results as with the competitive species models: a smaller time step can lead to a more accurate simulation of species population behavior. However, there is a more apparent difference between the Euler method and rk45 method in this case. The rk45 method actually appears to remain largely unaffected by the time steps, and it continues to show oscillatory behavior. Euler's method also doesn't vary too much with the smaller steps, but increasing the steps causes it to display strange results again. Just like with the competitive species, a time step that's too large sends both models off to some weird values.

Task 2

In this task, rather than vary the time step, I chose different values for the coefficients a, b, c, and d in the competitive species equations. The results for the different coefficient values for both methods are attached below. Note that Figure 1 and Figure 7 represent the values given in the lab manual for the competitive species and predator-prey model respectively, so I will omit those below and go straight for the varied coefficient graphs.

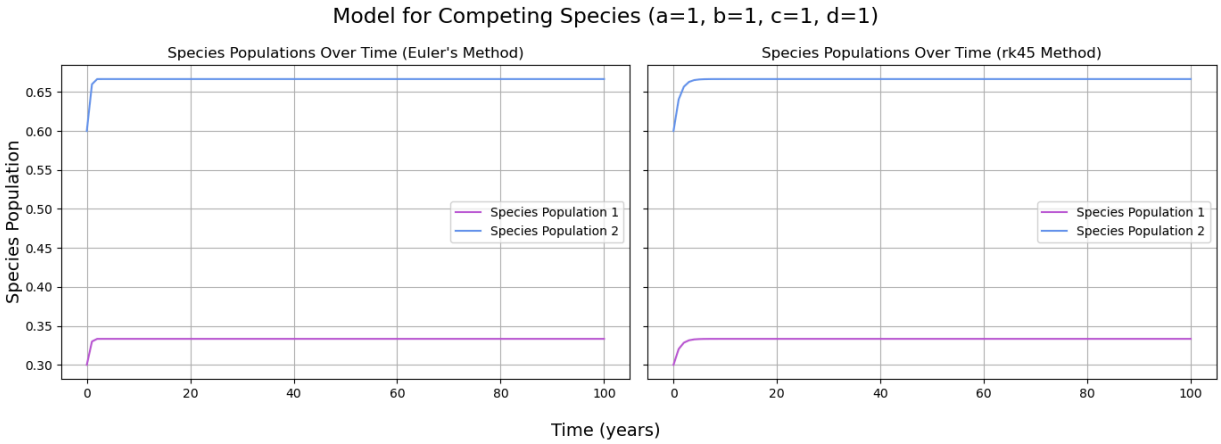


Figure 12

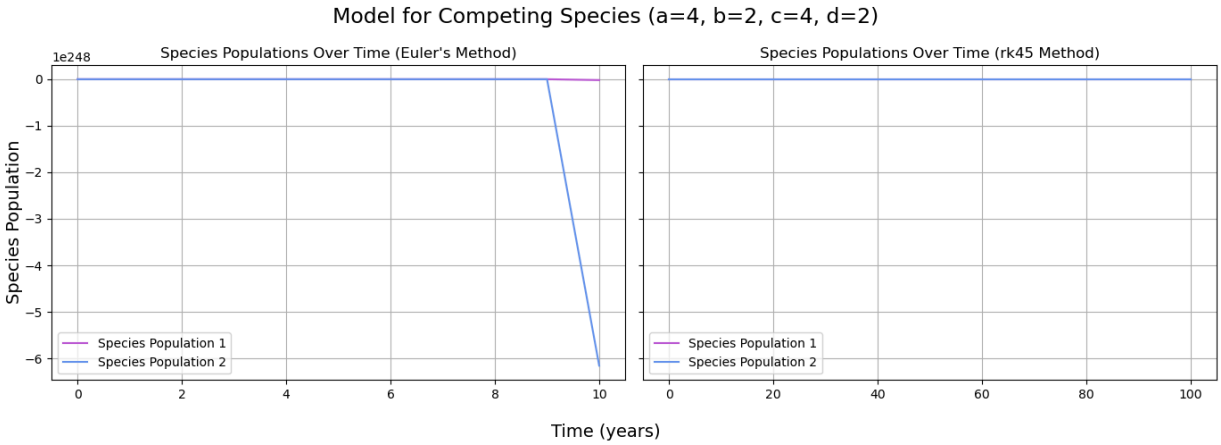


Figure 13

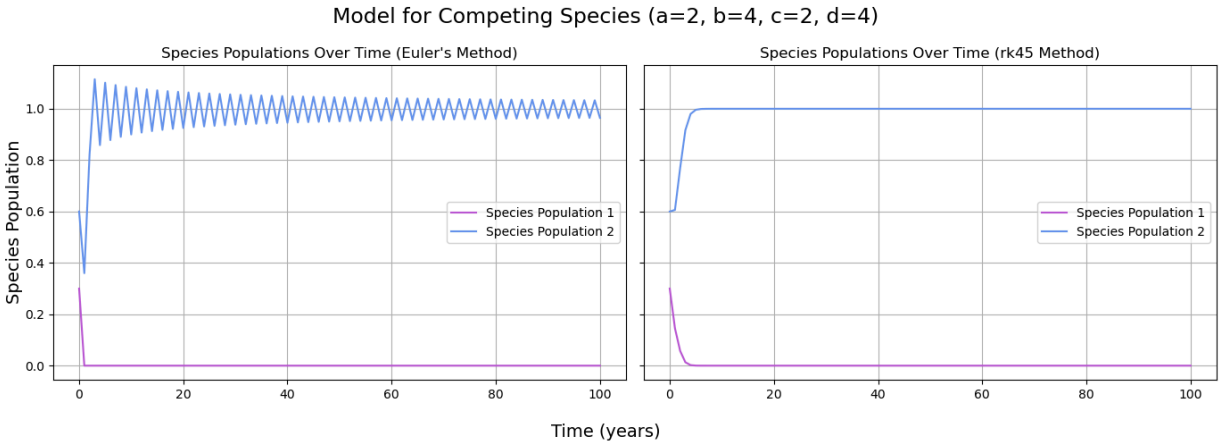


Figure 14

Figure 12 shows the results when a, b, c, and d all equal 1. Though the rk45 method has a slightly smoother curve, both rk45 and Euler's methods have the same graph. Both of them

model the two species growing for a bit after their initial population value, and then reaching a steady state fairly quickly. Next, in Figure 13 the resultant plots for a=c=4 and b=d=2 are shown. Having a higher value for a and c than b and d means the populations are more impacted by their own reproduction than they are by competition with the other species. The graph here displays odd results again, which could possibly be caused by coding error, or because the coefficients just skew the results that much. This looks similar to Figure 3 where these equations were tested with a time step of 3 years. To test further, I switched the values of a and c with the values of b and d. This also yielded interesting results. For both methods, the first species went to zero almost immediately. Then, Euler's method shows the second species going right to capacity, and then oscillating around that capacity with decreasing amplitude. The rk45 model simply brought the second species to capacity and held it there steady. This makes a little more sense, because increasing b and d means the species populations rely more heavily on each other's populations. Therefore, if one species starts to do really well, the other species will start to do very poorly.

Finally, I varied the initial populations for each species, as outlined in Figures 15-17. Note that all other parameters are kept to the values listed in the lab handout (a=1, b=2, c=1, d=3, and the time step is 1 year).

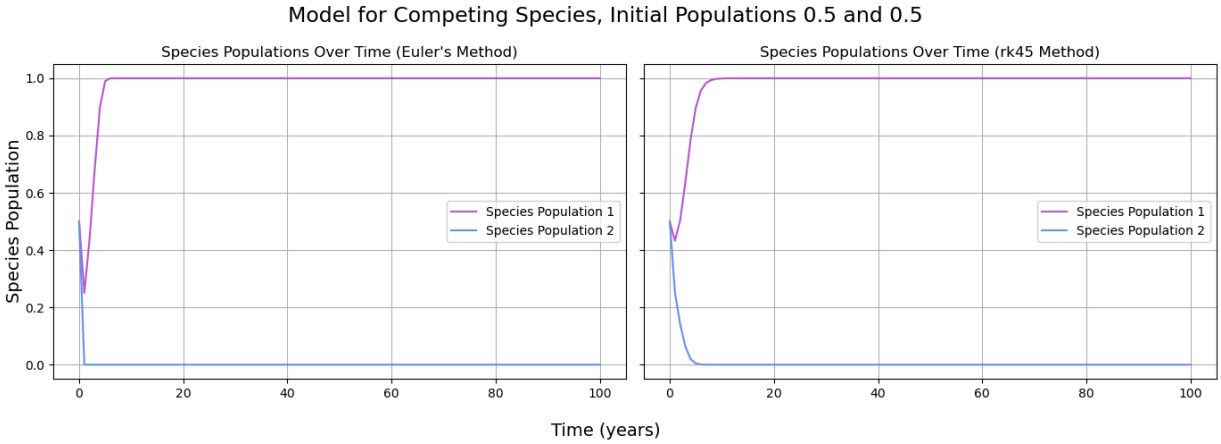


Figure 15

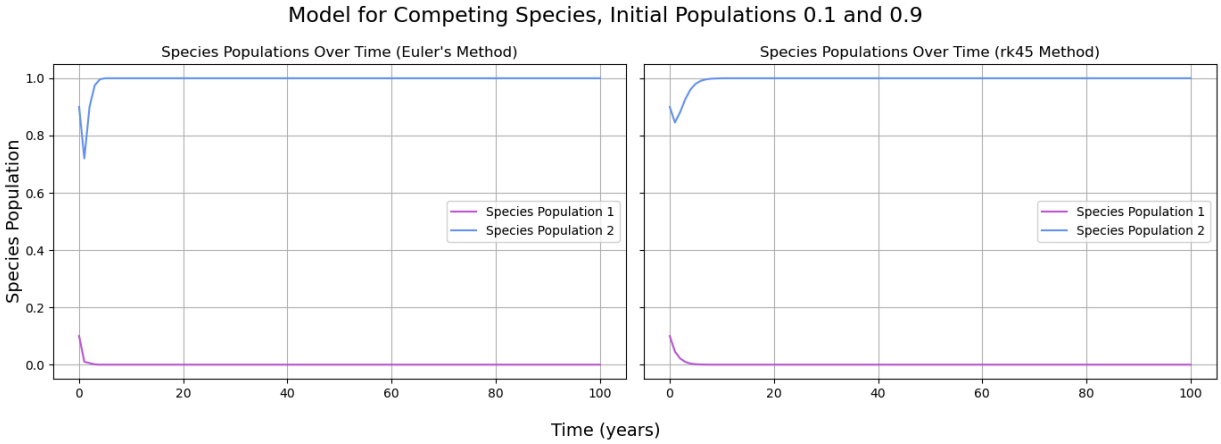


Figure 16

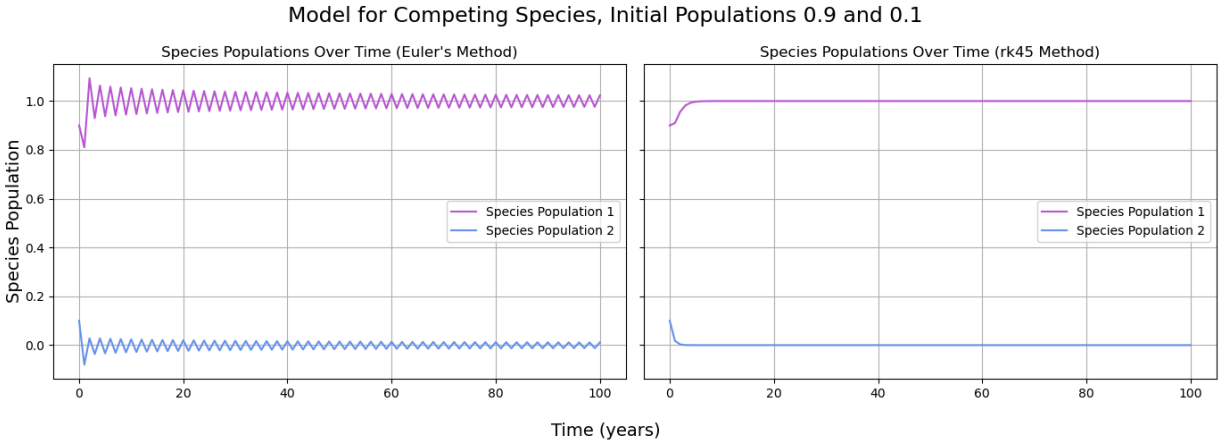


Figure 17

In Figure 15, each species starts at 0.5, and both models actually seem to agree for the most part on the behavior of each population with time. Species 1 takes a quick dip within the first couple years, then goes up to capacity, while the second species goes to zero fairly quickly. This behavior is also observed by both of the models in Figure 16 but with the species flipped and at different starting points: Species 1 starts at 0.1 and diminishes to zero, and Species 2 starts at 0.9, dips, then steadies at capacity. Figure 17, though, doesn't look as much like the other graphs. Euler's method once again displays oscillatory behavior around zero and capacity for Species 2 and 1 respectively. The rk45 method looks similar to the others, but the curves to the two extremes are smaller.

With all of this variation, it can be said that the coefficients in the Lotka-Volterra equations for competing species have a large impact on the behaviors of each species population with time. If the coefficients a and c are larger, they're going to increase the influence of reproduction on the species' populations; if b and d are larger, the size of the other species' population will have a greater impact on the other. Additionally, the starting population size is influential as well. When the two species start with the same population fraction, it will depend on the coefficient values to see which one stabilizes at capacity and which one goes to zero. If one species starts with a much higher population fraction than the other, it will go to capacity while the other diminishes to zero.

Task 3

For this task, I did the same procedure as Task 2, but this time using the predator and prey equations. I investigated the effects of different coefficients and initial populations using the same values as I did for Task 2. Additionally, I made phase diagrams for the original case with the given variable values in the lab manual. The results and corresponding discussions are below.

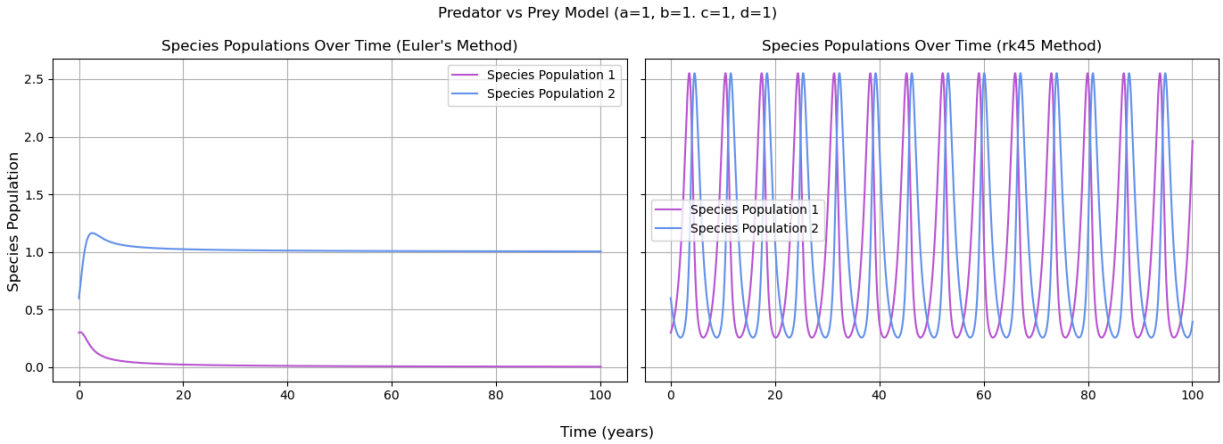


Figure 18

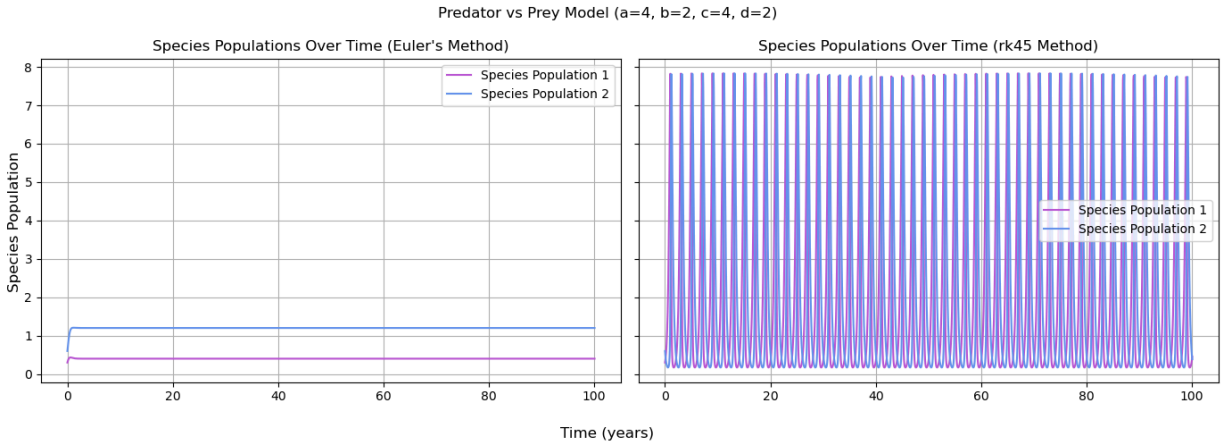


Figure 19

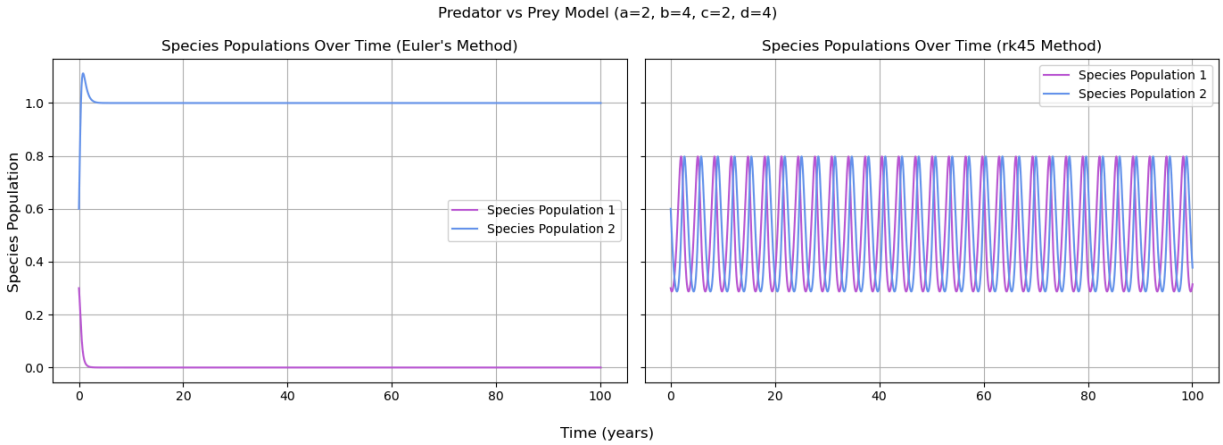


Figure 20

The different coefficients clearly lead to very different results. Note that Figure 7 depicts the behavior of the species populations with the values given in the lab manual, so that graph can

be used as a benchmark for comparison. Figure 18 shows what the populations would do when all the coefficients equal 1. In general, Euler's method looks similar to Figure 7, with the most apparent difference being a more gradual transition to each of the species equilibrium values. The rk45 method, though, has a much larger amplitude with the coefficients of 1. Both species also seem to oscillate around the same value, whereas in Figure 7 they were centered at different spots. With the coefficients of Figure 19, a similar thing happened once again, except this time the rk45 amplitude is even larger. Also, Species 1 didn't remain at 0, but rather had a slight increase and stayed there for the duration of the time period. In this scenario, the coefficients a and c are larger than b and d. This means that Species 1 (the prey) should have a greater population because their reproduction rate is large, and the death rate of Species 2 (the predators) is also large. However, the overall trend goes back and forth, because the larger population of prey means the predators have a more abundant food source. On the flip side of this, when a and c are smaller than b and d as shown in Figure 20, the results also look like Figure 7. However, the rk45 method results in a graph with higher amplitude and higher frequency. In this context, b and d being larger means the prey is being hunted more, decreasing their population. This also means the food availability for the predators decreases, which has a negative effect on their numbers as well.

The next step is to vary the initial populations of each of the species. Again, I used the same values as I did for Task 2, and the results are attached below.

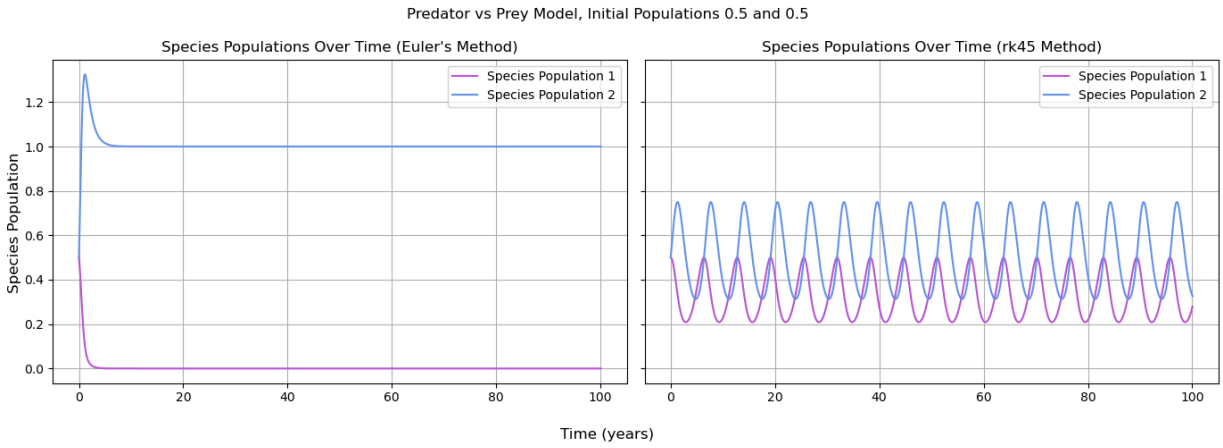


Figure 21

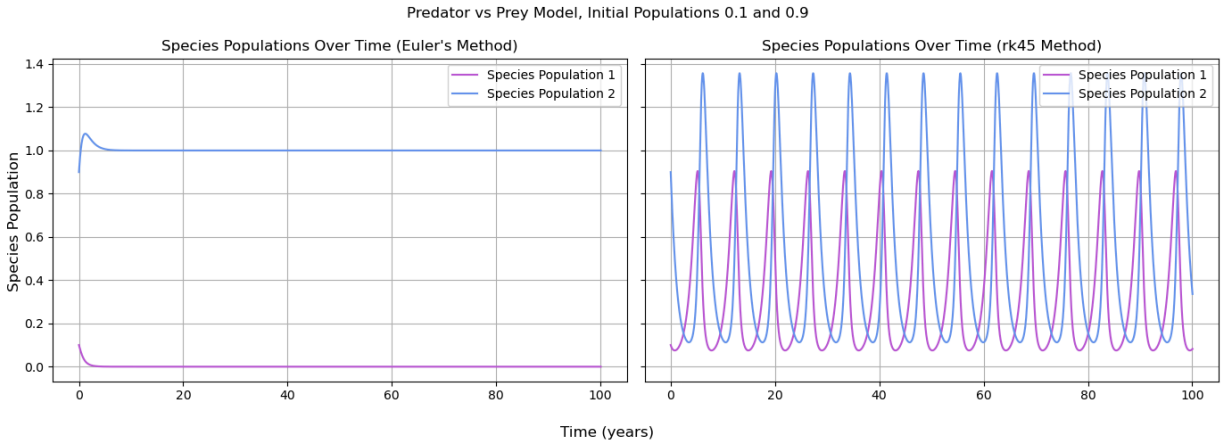


Figure 22

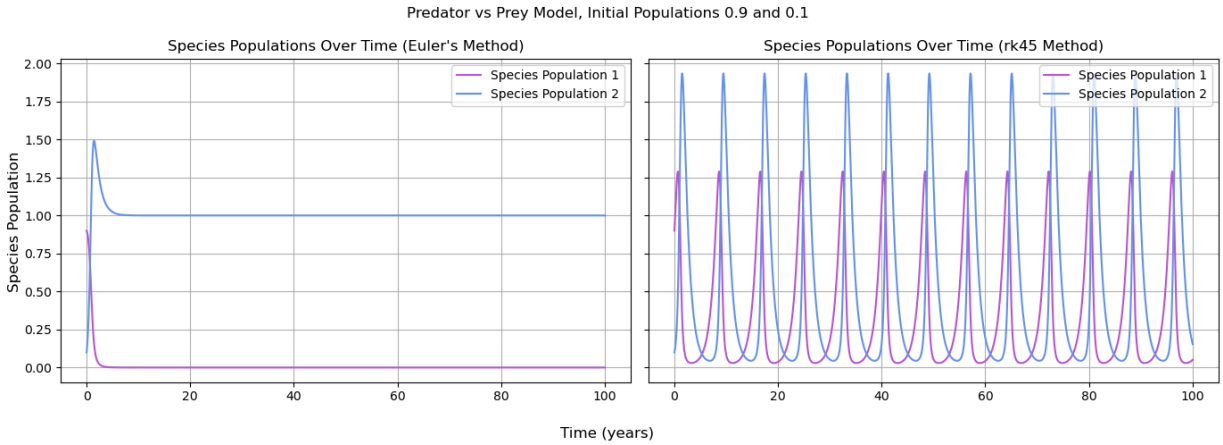


Figure 23

In all three of these variations, the graphs generally look quite similar. For all the Euler's methods, there seems to be a consensus that the predator population sees a bump in numbers and then stabilizes at capacity quickly thereafter. Additionally, the prey population heads straight for zero from the start. For the rk45 methods, the amplitude changes, but the phase and frequency looks to be about the same. The highest amplitude is found when the prey starts at 0.9 capacity and the predators start with 0.1, and the smallest is when they're both at 0.5 initially. Based on these graphs, we can see that having more prey can help the predator population grow because it indicates more food availability. However, when the predator population grows, the prey population decreases with more demand for food. Decreasing the prey population decreases the predator population, but then that allows prey more time to reproduce, thereby increasing themselves, and the cycle repeats (hence the oscillatory nature demonstrated by the rk45 methods).

Finally for this task, I made a phase diagram with the prey population on the x-axis and predator population on the y-axis. This kind of graph is helpful because it can show irregular behavior in the populations. The result is attached below.

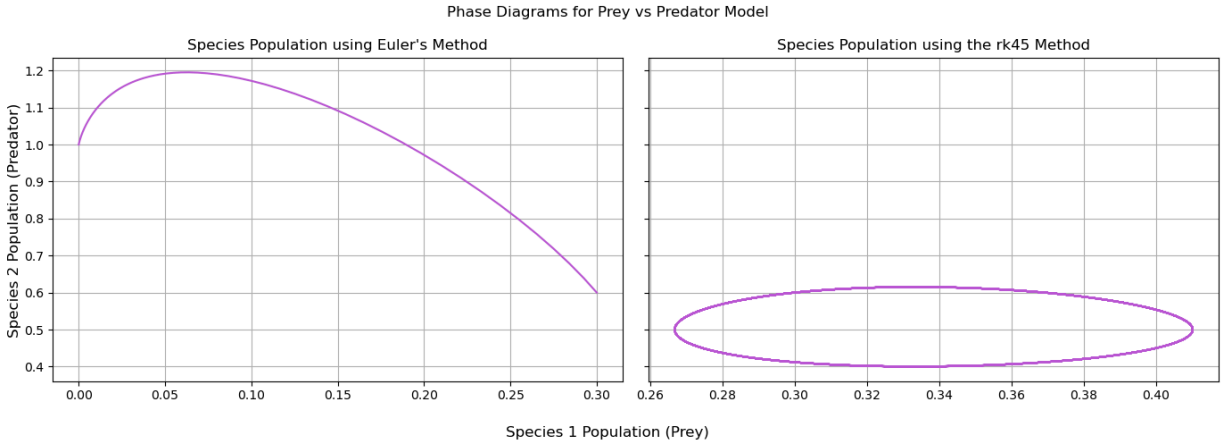


Figure 24

From these phase diagrams, clear differences are evident between Euler's method and the rk45 method. Euler's method shows a tendency for the populations to either stabilize at a single value or face extinction, while rk45 shows an infinite oscillatory cycle. Plotting the data in this way allows us to see the overall trend or nature of the populations based on the equations given with certain specified parameters.

Task 4

For the final task of this lab, the rk45 method was modified from its Lotka-Volterra applications to investigate atmospheric behavior. This time, the set of three Lorenz equations with its given coefficient values were implemented into the function. This task is similar to the others in that the goal is to investigate how changing the parameters affects the final results.

First, I looked at the Lorenz equations and compared the results when r=23 and r=25. I simultaneously investigated their respective behavior with perturbations (a small increment of 10^-5) added as well. The important thing to look for is where the blue and purple lines line up and where they differ. The times they line up indicate more predictable behavior, while the times they are not in line indicate times of very low predictability. The plot of these results is below.

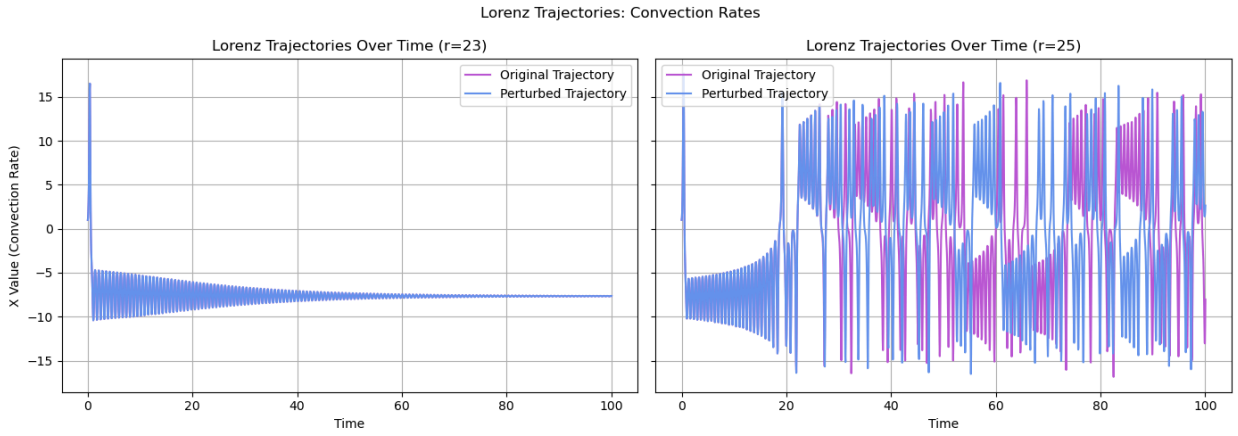


Figure 25

Next, I varied the initial conditions σ and b to see how the results changed for r=23, r=25, and both perturbed as well. First, I changed σ to a lower and a higher value. Then I changed b to a higher and lower value than the one given as well. All these results are attached below in Figures 26-29.

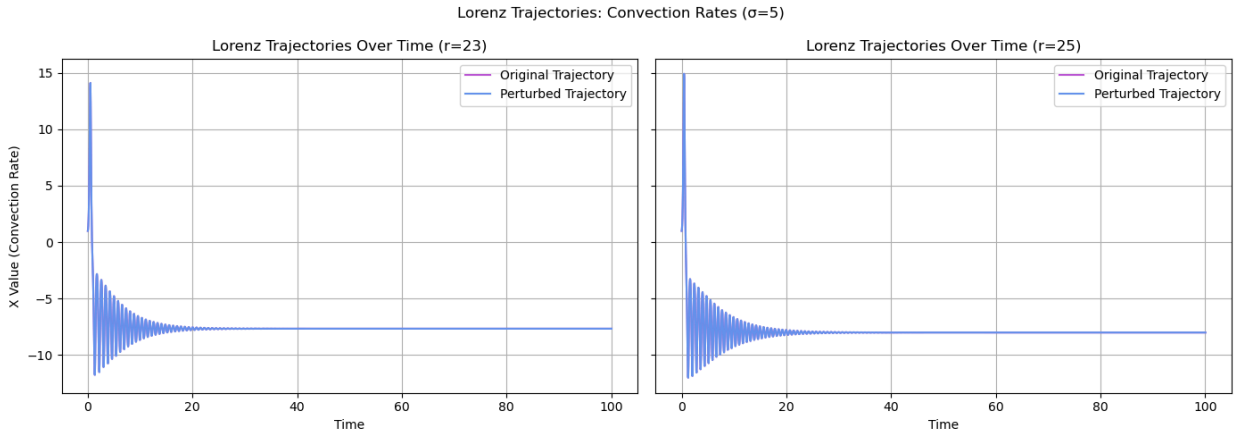


Figure 26

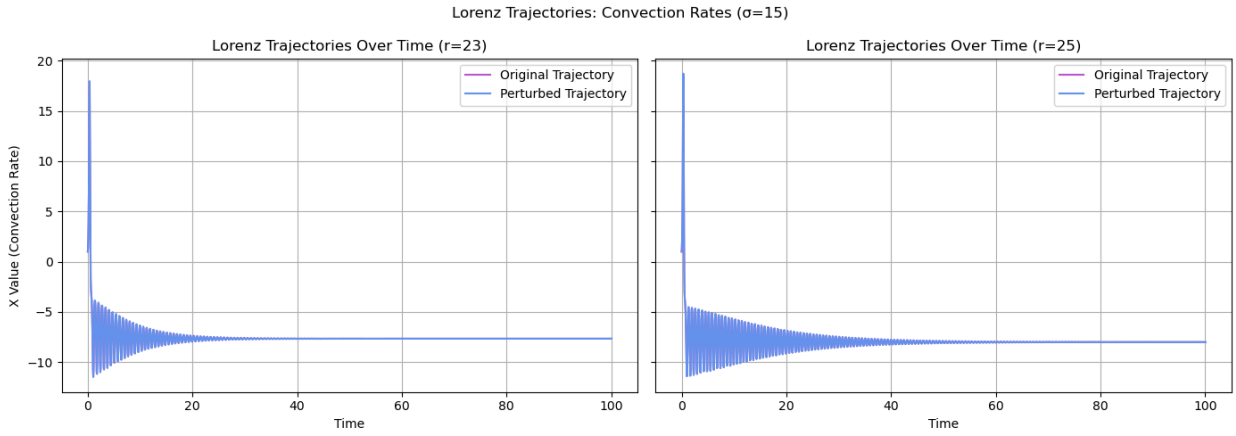


Figure 27

In Figures 26 and 27, I tried varying the value of σ to be 5 and 15. In both cases, there is no difference between the original and perturbed trajectories, which means the predictability is pretty high. Based on these results here, it can be said that σ variations are able to improve predictability.

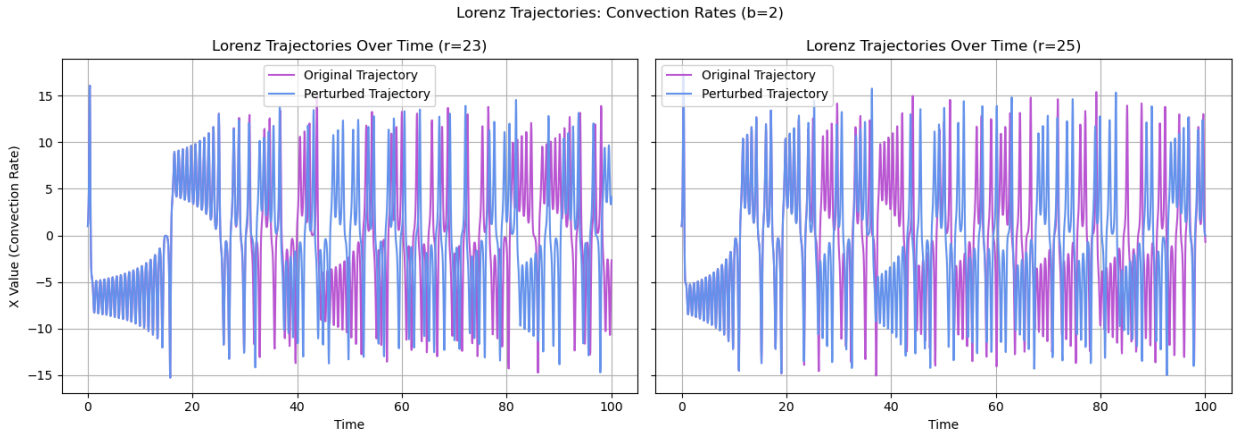


Figure 28

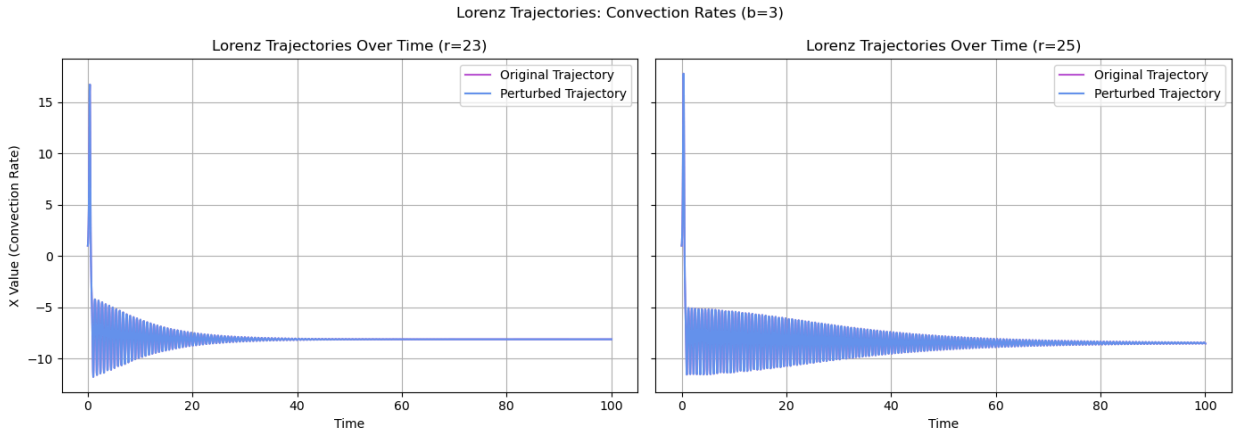


Figure 29

In Figures 28 and 29, I varied the value of b to be 2 and 3, which is just under or over the given value. When b=2, the predictability very clearly dropped, especially for r=23. However, when b=3, the predictability seemed to improve. Some other differences, though, include the amplitude and dampening effect on the functions. For r=25 this is quite notable, as the amplitude was much lower than in Figure 25, and dampens more gradually than any of the other graphs. This corresponds to the weather itself: higher amplitudes indicate larger swings in weather intensity, and the dampening just shows how those swings decrease over time. With a slightly higher b value, the intensity of weather systems had an apparent increase, and it took longer for that intensity to decrease.

What all of these results show is that it is much harder to create an accurate weather prediction the farther you go out in time. Predictability is generally much higher for nonlinear systems in shorter timeframes, as shown in the graphs. Since weather is a nonlinear system, that principle applies. This is evident in the real world, as it is common to see forecasts come true if they're predicting the weather within a couple of days; however, if you were to look at a forecast for a date a week from today, you'd notice it changes as it gets closer to that date.